

Clostridium difficile infection in humans and piglets: a 'One Health' opportunity.



Clostridium difficile causes infectious diarrhoea in humans and animals. It has been found in both diarrhoeal and non-diarrhoeal pigs, horses and cattle, suggesting a potential reservoir for human infection, and in 20-40 % of meat products in Canada and the USA, suggesting the possibility, albeit not proven, of food-borne transmission. Although it is not yet completely clear, it is likely that excessive antimicrobial exposure is driving the establishment of *C. difficile* in animals, in a manner analogous to human infection, rather than the organism just being normal flora of the animal gastrointestinal tract. PCR ribotype 078 is the most common ribotype of *C. difficile* found in pigs (83 % in one study in the USA) and cattle (up to 100 %) and this ribotype is now the third most common ribotype of *C. difficile* found in human infection in Europe. Human and pig strains of *C. difficile* are genetically identical in Europe confirming that a zoonosis exists. Rates of community-acquired *C. difficile* infection (CDI) are increasing world wide, a fact that sits well with the notion that animals are a reservoir for human infection. Thus, there are three problems that require resolution: a human health issue, an animal health issue and the factor common to both these problems, environmental contamination. To successfully deal with these recent changes in the epidemiology of CDI will require a 'one health' approach involving human health physicians, veterinarians and environmental scientists.